

4 - ...
 $\Rightarrow$ Linear Regression (one independent
one dependent)

Predict Sales based on TV Advertisement Budget

TV Ads (cost)	Sales (y)
10	15
20	25
30	35
40	45
50	55

Task: 1. Prediction for 55, 60, 20 TV Ad budget.
2. Plot.

Formulas:

$$y = ax + b$$

$$a = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$b = \bar{y} - a\bar{x}$$

let, $\bar{x} \rightarrow$ TV Ads, $\bar{y} =$ Sales

$$\bar{x} = \frac{(10+20+30+40+50)}{5} = 30.0$$

$$\bar{y} = \frac{(15+25+35+45+55)}{5} = 35$$

$$a = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{800}{800} = 1$$

x	y	(x - $\bar{x}$)	(y - $\bar{y}$)	(x - $\bar{x}$) × (y - $\bar{y}$)	(x - $\bar{x}$) ²
10	15	-20	-20	400	400
20	25	-10	-10	100	100
30	35	0	0	0	0
40	45	10	10	100	100
50	55	20	20	200	200
				<u>800</u>	<u>800</u>

$$a = \frac{800}{800} = 1$$

$$b = \bar{y} - a\bar{x}$$

$$b = 35 - (1)(30)$$

$$b = 5$$

$$\boxed{a = 1, b = 5}$$

$$y = ax + b$$

$$y = 1x + 5$$

To find, $x = 55, x = 60, x = 20$

$$y = 55 + 5 = 60 \Rightarrow$$

$$\text{TV Add (55)} = \text{Sale (60)}$$

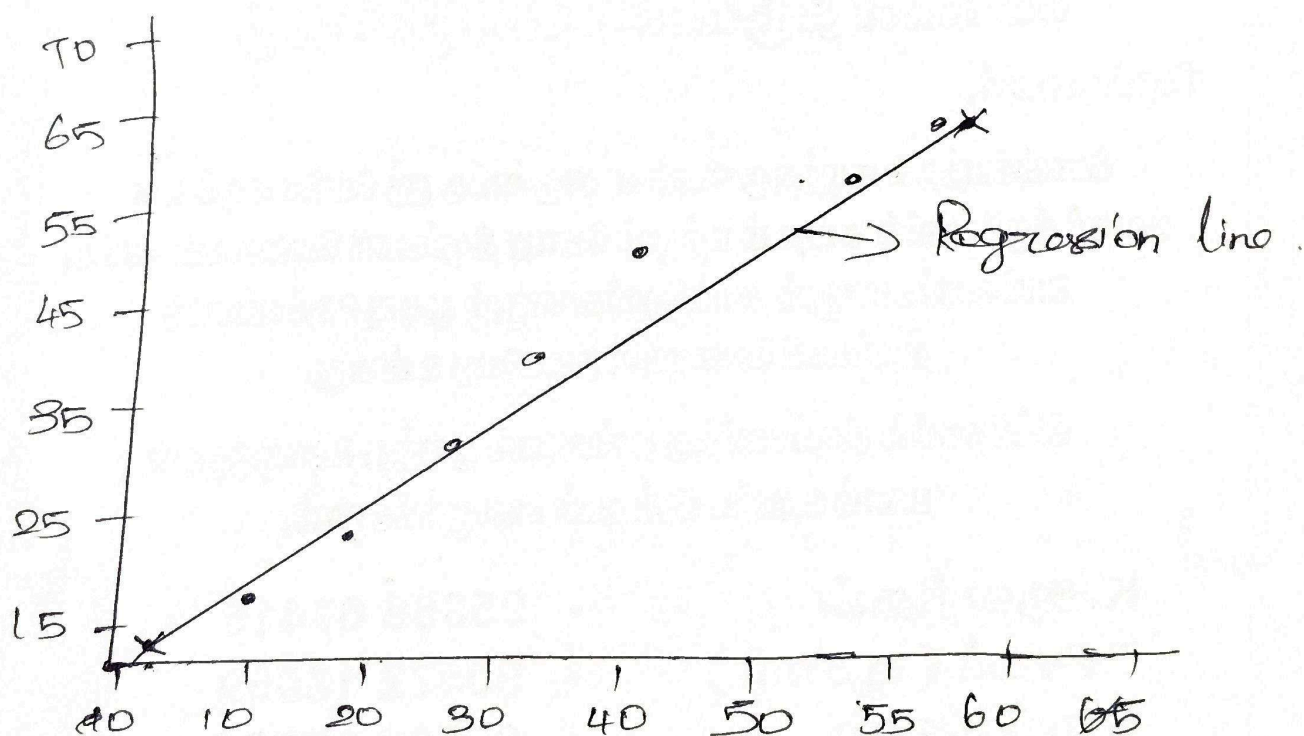
$$y = 60 + 5 = 65 \Rightarrow$$

$$\text{TV Add (60)} = \text{Sale (65)}$$

$$y = 20 + 5 = 25 \Rightarrow$$

$$\text{TV Add (20)} = \text{Sale (25)}$$

Plot



Regression line (Two points) let $x = 1, x = 60$

$$y = 1 + 5 = 6 \quad (1, 6)$$

$$y = 60 + 5 = 65 \quad (60, 65)$$

} straight line